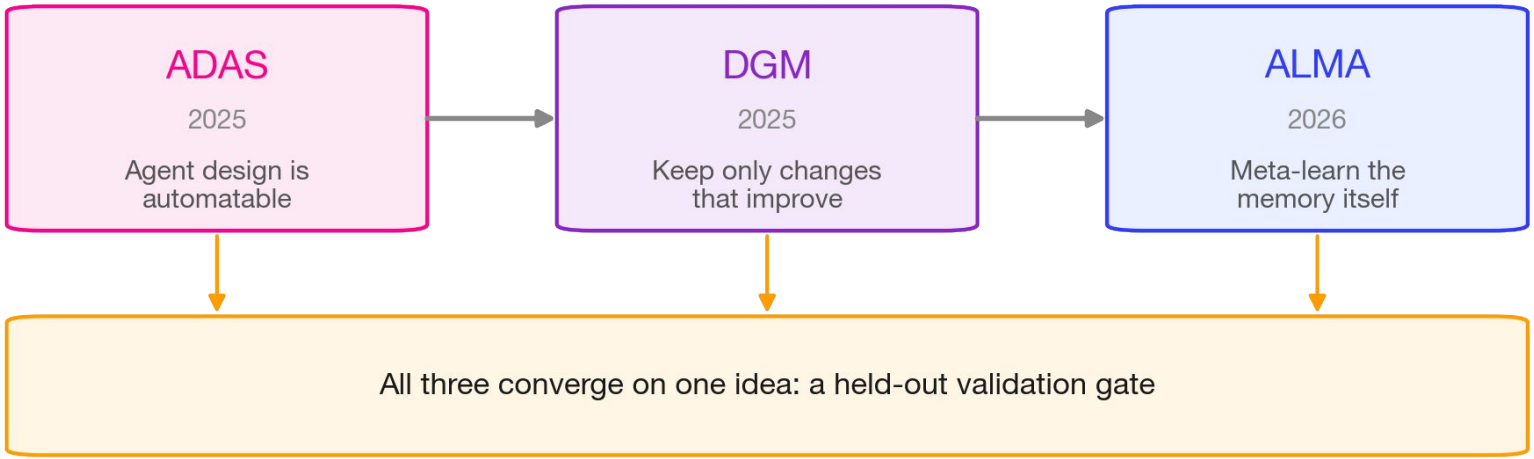
DAY 2 · SESSION 3

# Self-improving Agentic Systems

*Where does the adaptive agent fit in the research landscape?*

VECTOR INSTITUTE | INSTITUT VECTEUR

# Three papers, one arc




*ADAS → DGM → ALMA: each builds on the last — all share a held-out gate.*

# ADAS → DGM: two ideas that transfer

### ADAS

Meta-agent writes agents as code

Discovers designs beating hand-crafted ones

Our curriculum: a constrained version

### DARWIN GÖDEL MACHINE

Self-rewriting + an empirical gate

20% → 50% on SWE-bench, one agent

Archive of diverse solutions

**We have a version of the gate (graduate_hypothesis). We don't have the archive.**

# ALMA learns the memory design itself.

Statelessness is the bottleneck. Schema, retrieval, and update rules — meta-learned from task performance, not hand-crafted.

> Our adaptive agent hand-designed what ALMA would learn.

# What ALMA means for forecasting

*Three memory components we hand-designed — and where the frontier takes each*

| Memory part | We hand-designed | The ALMA frontier |
|---|---|---|
| **Schema** | obs → hyp → calib → narrative | Meta-learned from performance |
| **Retrieval** | Full strategy, every prediction | Filtered by regime + horizon |
| **Update rules** | Evidence bars + threshold | Validated out-of-sample |

**The update rules are the most actionable gap — and the bridge to track two.**

# The hand-designed schema

```python
class WtiStrategyState(AdaptiveSkillState):
    approach_narrative: str
    calibration_corrections: list[...]
    hypotheses: list[Hypothesis]
    observations: list[Observation]
    version_history: list[VersionEntry]
```

*adaptive_agent/skill_state.py*

## This schema is ours.

Five typed fields, an evidence hierarchy — designed in one sitting for WTI.

- Deliberate and sound
- But not discovered
- ALMA would search over it

# The honest lineage

*What we have from each paper — and the gap that remains*

| Paper | What we have | The gap |
|---|---|---|
| **ADAS** | Curriculum writes text | No code search |
| **DGM** | Confirmation gate | No archive |
| **ALMA** | Fixed schema + rules | No adaptive retrieval |
| **SkillOpt** | Curriculum self-edits | No held-out gate |

**The held-out gate is the gap that matters most.**

A SECOND TRACK

# Skills as trainable state

*SkillOpt and the discipline of text-space optimization*

VECTOR INSTITUTE | INSTITUT VECTEUR

# SkillOpt: skill as trainable state

### The artifact

A markdown skill file as the frozen agent's trainable state — our wti-strategy is exactly this.

### The optimizer

A separate model turns scored rollouts into bounded edits — like DSPy, but for skill docs.

### The gate

Each edit is accepted only if it improves a held-out window; rejected edits become feedback.

**+23.5 pt average gain over no-skill (GPT-5.5, six benchmarks) — held-out gated.**

VECTOR INSTITUTE | INSTITUT VECTEUR

# The shared principle

01

## DGM

**accept / reject**

Keep an edit only on evidence of improvement on a held-out benchmark.

02

## ALMA

**out-of-sample**

Update rules judged on task performance the design never trained against.

03

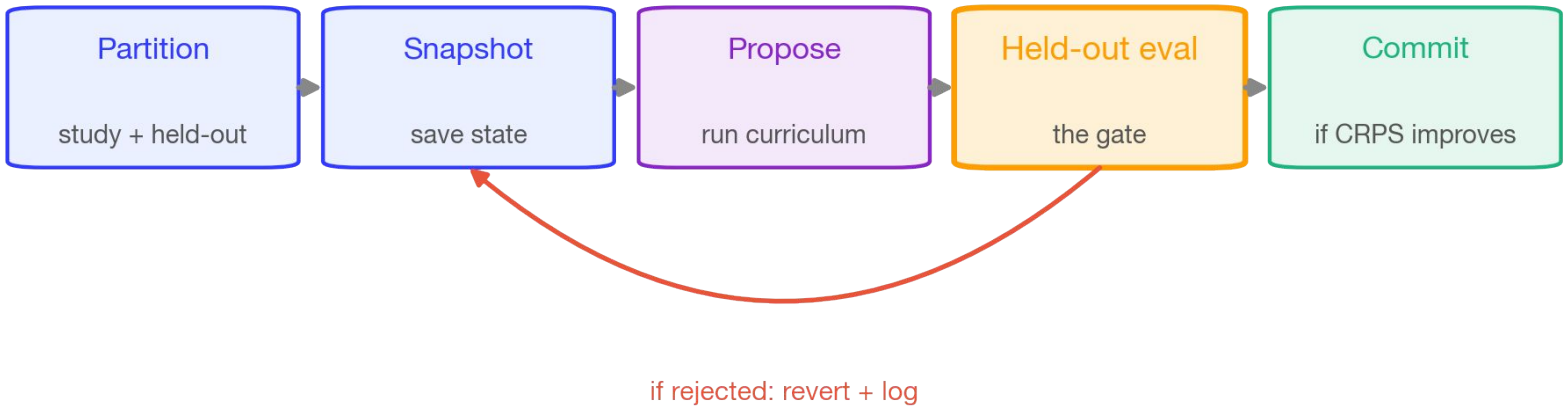
## SkillOpt

**held-out CRPS**

Commit a skill edit only if it improves an unseen validation window.

**Change is easy. Improvement requires a held-out gate.**

# Validation-gated curriculum




*Inspired by, not replicating: every committed change must first improve an unseen window.*

# Add one held-out gate.

That's the principle ADAS, DGM, ALMA, and SkillOpt share — applied to the curriculum you already have.

> The evaluation harness is already there. The question is yours.